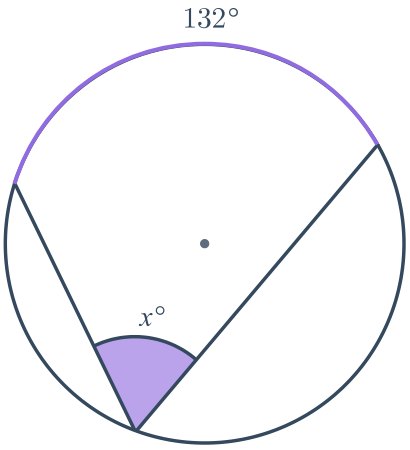
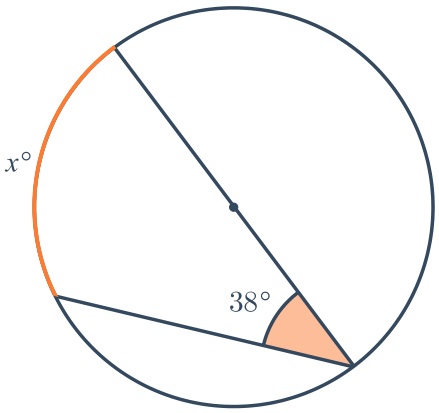
Inscribed angles and polygons

1 Solve for value of the variable(s) in the following diagrams.

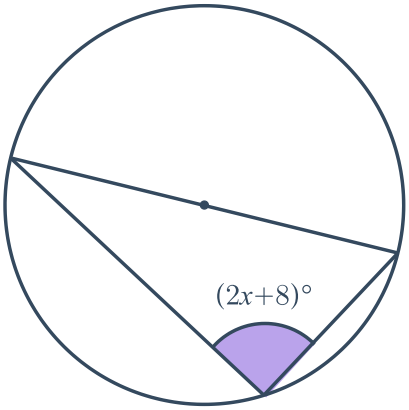
a



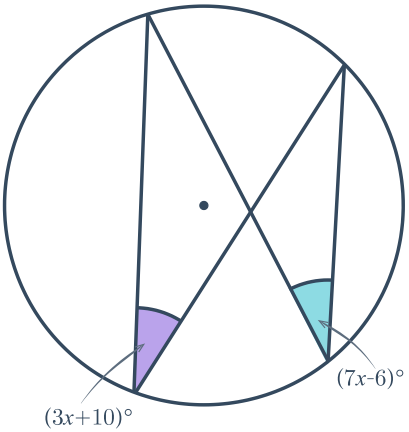
b



c

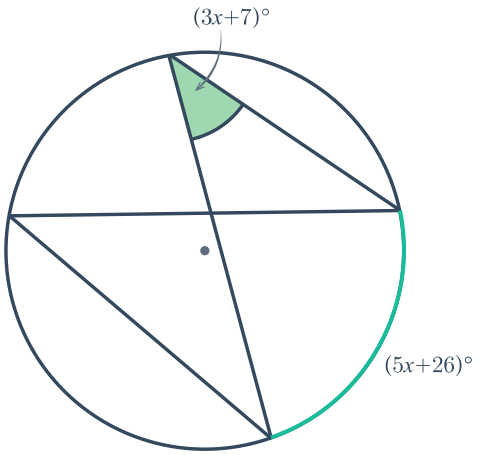


d

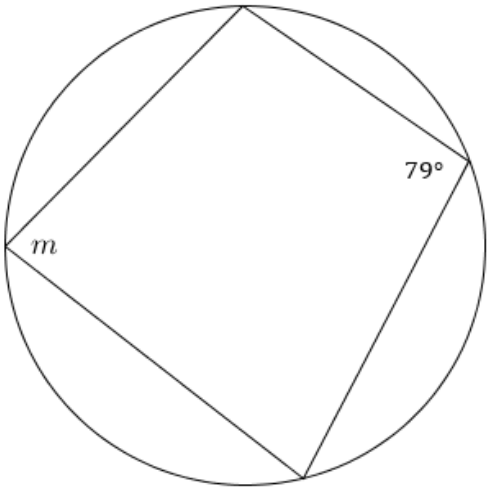


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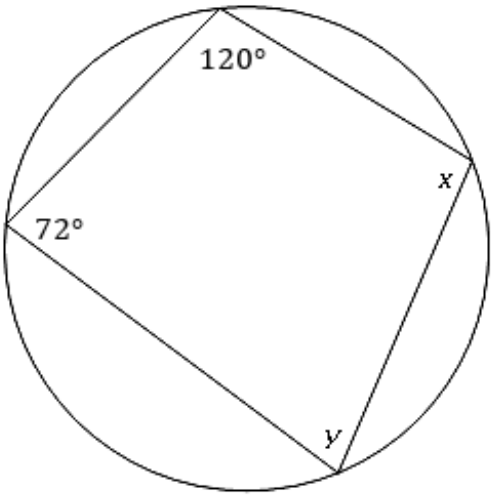
e



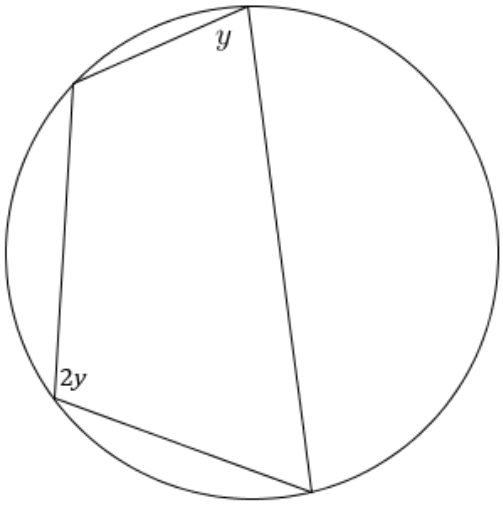
f



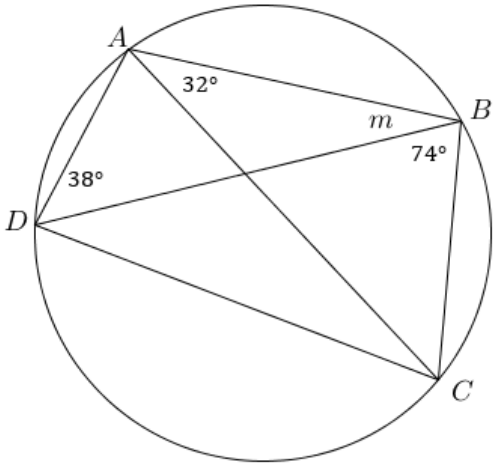
g



h



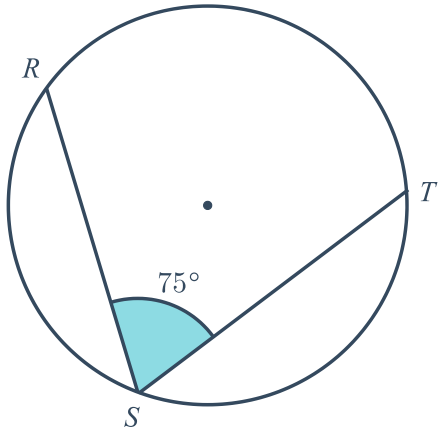
i



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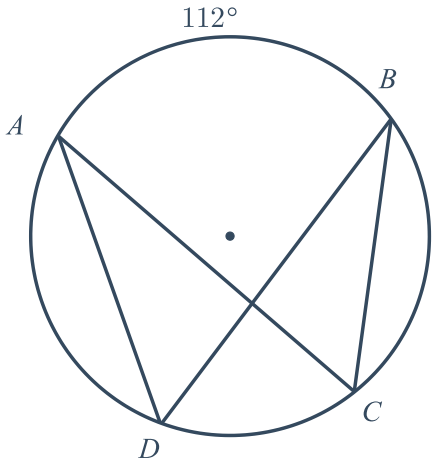
2 Consider the following diagram.

Find the measure of the minor arc $\overset{\frown}{RT}$.



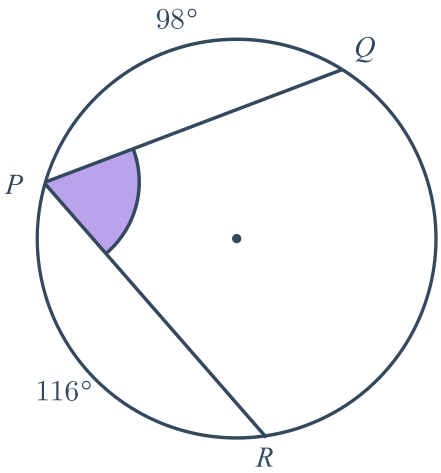
3 Consider the following diagram.

Find $m\angle ADB$.



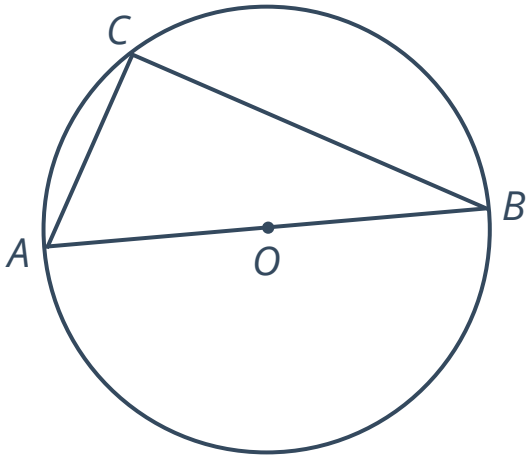
4 Consider the following diagram.

Find $m\angle RPQ$.



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5 Consider the following diagram where A, B, and O are collinear.
Find $m\angle ACB$.

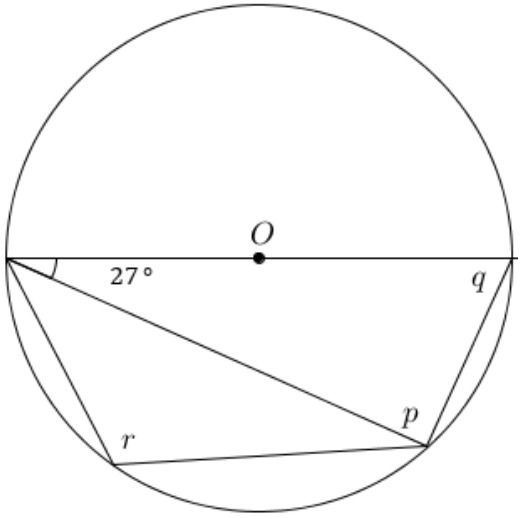


6 In the diagram, O is the center of the circle. A quadrilateral inscribed in the circle and one of its diagonal are drawn.

a Solve for p.

b Solve for q.

c Solve for r.

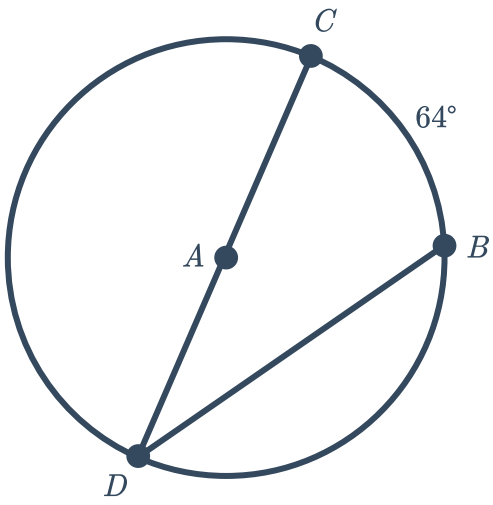


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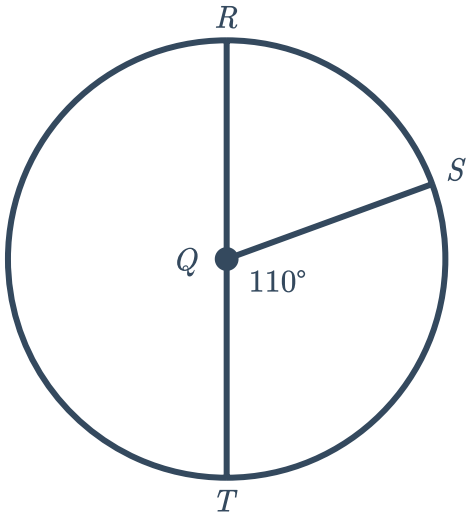
Additional questions

7

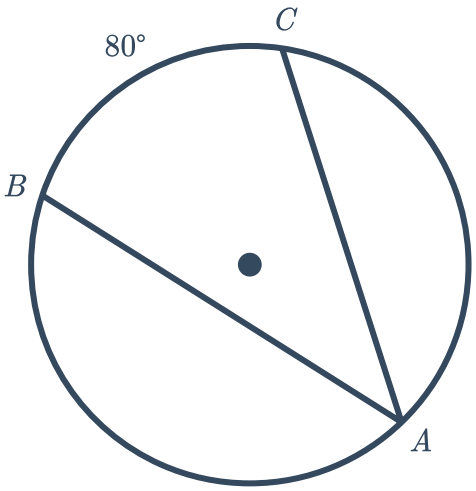
a Solve for $m\angle CDB$.



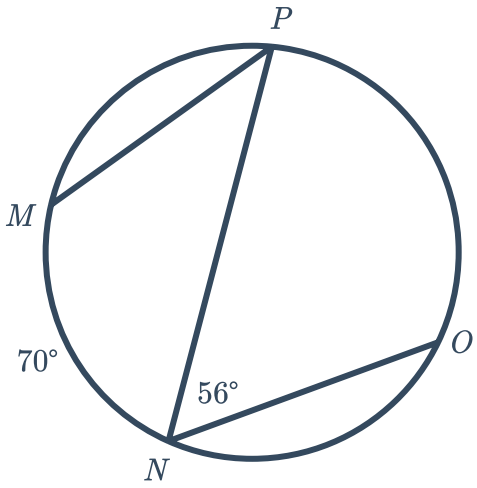
b Solve for $m\overset{\frown}{RS}$.



c Solve for $m\angle BAC$.

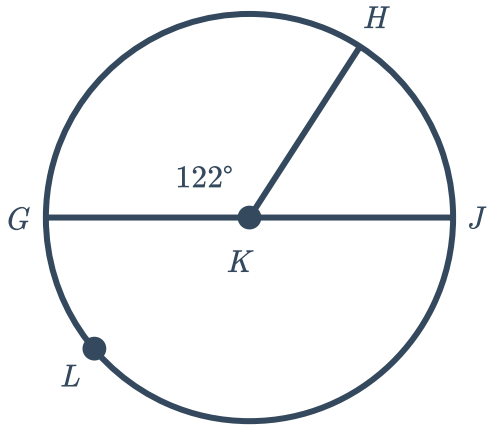


d Solve for $\angle MPN$.

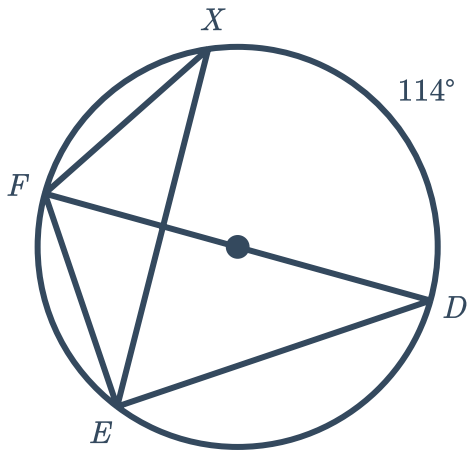


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e Solve for $m\overset{\frown}{GH}$.

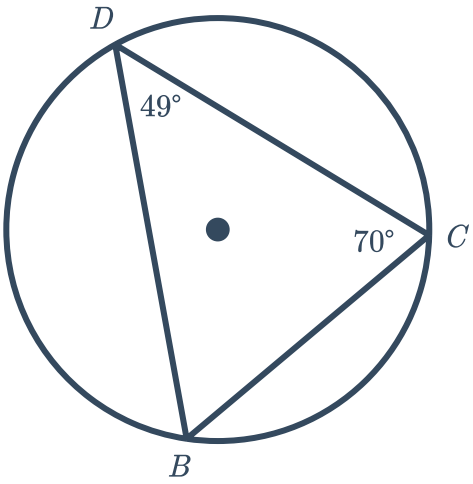


f Solve for $m\angle XFD$.

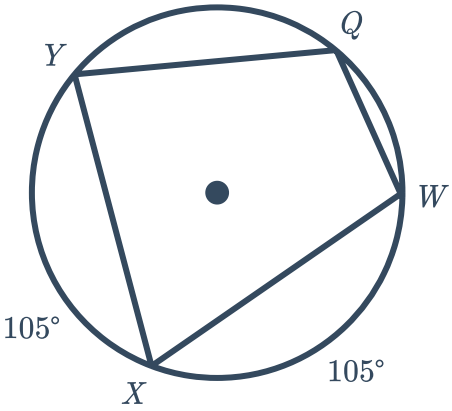


8

a Solve for $m\overset{\frown}{DC}$.

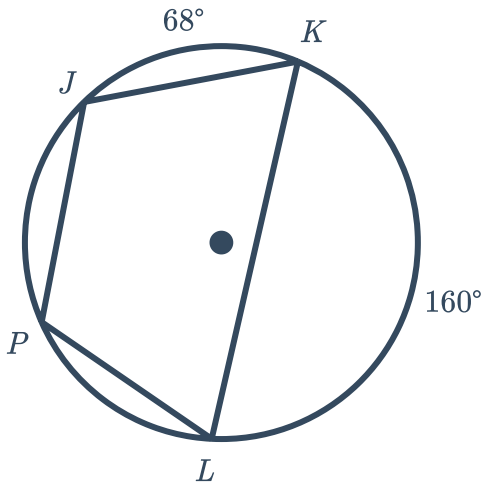


b Solve for $m\angle Q$.



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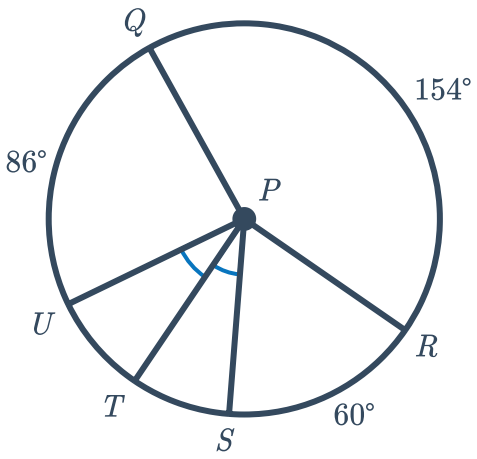
9



a Solve for $m\angle P$

b Solve for $m\angle K$

10

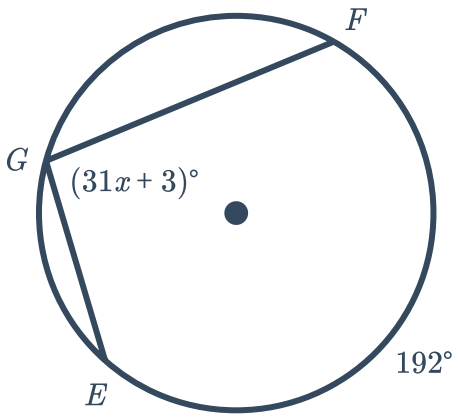


a Solve for $m\angle UPT$

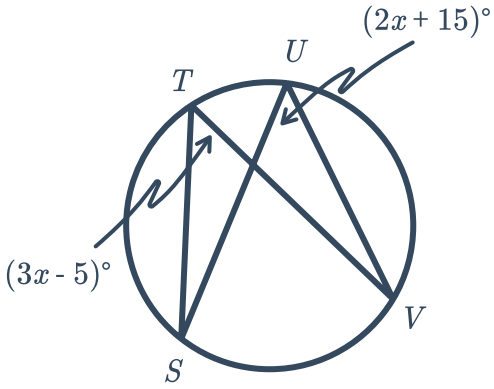
b Solve for $m\overset{\frown}{US}$

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11 Solve for x



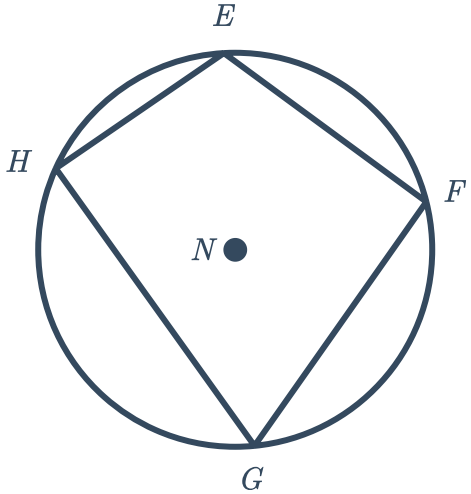
12



a Solve for x

b Solve for $m\overset{\frown}{SV}$

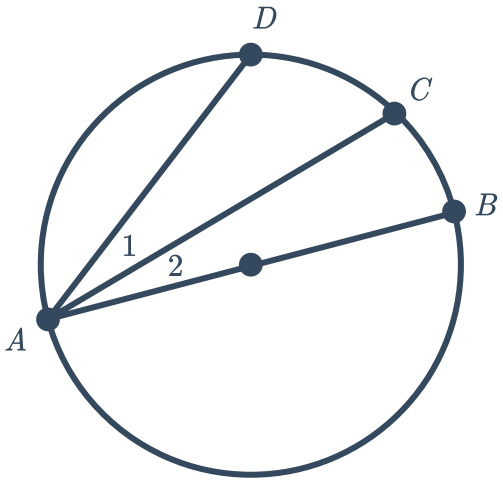
13 If $m\angle H = a^\circ$, what is $m\angle F$?



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14 Given:

- The ratio of $m\overset{\frown}{BC} : m\overset{\frown}{BD}$ is $2 : 5$.
- $m\angle 2 = 24^\circ$



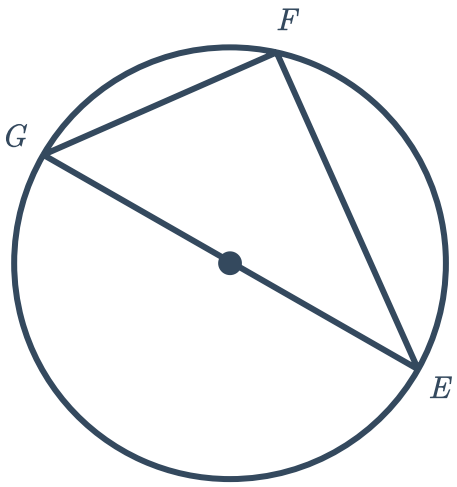
a Solve for $m\angle 1$

b Solve for $m\overset{\frown}{BD}$

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15 Given:

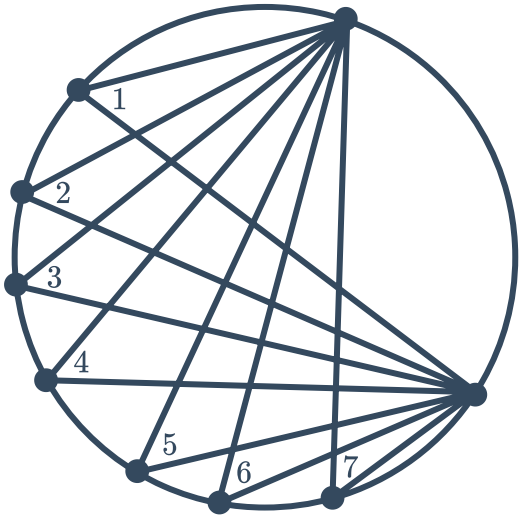
- $\overline{GE}$ is a diameter of the circle
- $m\overset{\frown}{FE} = (9x + 18)^\circ$
- $m\angle G = 34^\circ$



a Solve for x

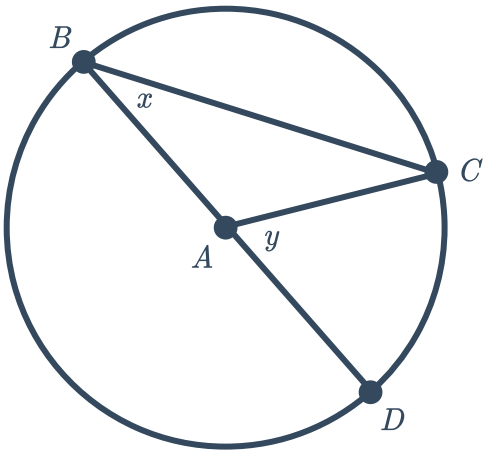
b Solve for $m\overset{\frown}{GF}$

16 What is the relationship between the inscribed angles in the circle? Justify your answer.

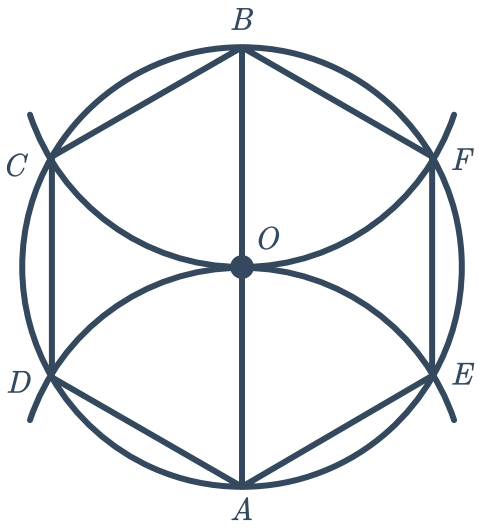


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17 Show that $y = 2x$.



18 Andie constructed a regular hexagon inscribed in circle O:



a What is the measure of each angle formed by two sides of the hexagon?

b List all of the arcs with the same measure as $\overset{\frown}{CDA}$.

c What is the relationship between $m\overset{\frown}{COF}$ and $m\overset{\frown}{CAF}$?

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